

◆ Arithmetic Operators (+ − * / % **)

Q1

What is the output?

```
console.log(10 % 3);
```

Q2

```
console.log(2 ** 3 ** 2);
```

Q3

```
let x = 5;  
console.log(x++ + ++x);
```

Q4

```
console.log("5" - 2);  
console.log("5" + 2);
```

Q5

What is the result?

```
console.log(10 / 0);
```

◆ Assignment Operators (=, +=, -=, *=, /=)

Q6

```
let x = 10;  
x += 5 * 2;  
console.log(x);
```

Q7

```
let a = 5;  
a *= 2 + 3;  
console.log(a);
```

Q8

Predict output:

```
let x = 4;  
x = x += 3;  
console.log(x);
```

◆ Comparison Operators (==, ===, !=, !==, >, <)

Q9

```
console.log(5 == "5");  
console.log(5 === "5");
```

Q10

```
console.log(null == undefined);  
console.log(null === undefined);
```

Q11

```
console.log(false == 0);  
console.log(false === 0);
```

Q12

```
console.log("2" > 1);  
console.log("02" == 2);
```

◆ Logical Operators (&&, ||, !)

Q13

```
console.log(true && false || true);
```

Q14

```
console.log(0 || "Hello");  
console.log("" && "World");
```

Q15

```
let x = 10;  
console.log(x > 5 && x < 20);
```

Q16 (Tricky)

```
console.log(!"");  
console.log(!0);  
console.log(!null);
```

◆ Bitwise Operators (&, |, ^, ~, <<, >>)

Q17

```
console.log(5 & 1);
```

Q18

```
console.log(5 | 1);
```

Q19

```
console.log(5 ^ 1);
```

Q20

```
console.log(5 << 1);
```

◆ Ternary Operator (? :)

Q21

```
let age = 18;  
let result = age >= 18 ? "Adult" : "Minor";  
console.log(result);
```

Q22

```
let x = 5;  
console.log(x % 2 === 0 ? "Even" : "Odd");
```

◆ Type Operators (typeof, instanceof)

Q23

```
console.log(typeof null);
```

Q24

```
console.log(typeof NaN);
```

Q25

```
console.log([] instanceof Array);  
console.log([] instanceof Object);
```

◆ Special / Tricky Interview Questions

Q26

```
console.log([] + []);
```

Q27

```
console.log([] + {});
```

Q28

```
console.log({} + []);
```

Q29

```
console.log(true + true);
```

Q30

```
console.log("10" - "4" - "3");
```

🔥 Scenario-Based Questions

Q31

Write a condition using logical operators:

- Age between 18 and 60
 - Salary greater than 20,000
-

Q32

Write a program using ternary operator:

- Check if number is positive, negative, or zero.
-

Q33

Explain difference between:

- ==
- ===